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Paleoposition of Southwest Japan at 16 Ma: implication from paleomagnetism of the Miocene Ichishi Group

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New paleomagnetic data from shallow-marine sediments of the Ichishi Group suggest a clockwise tectonic rotation of Southwest Japan in the Middle Miocene. Samples have been collected from mud or tuff layers at 17 sites. Stability of remanent magnetization has been examined by using alternating field and thermal demagnetization. The polarity sequence, composed of four normal and seven reversed polarity sites, is correlated to Polarity Epoch 16 (15.2–17.6 Ma), based on micropaleontological assignment of the upper Ichishi Group to Blow's Zone N8. The mean paleomagnetic direction of the 11 sites shows an anomalous declination toward the northeast. This result suggests that Southwest Japan was subjected to a clockwise rotation through 45° since 16 Ma. The clockwise rotation can be explained by the drift of Southwest Japan associated with the spreading of the Japan Sea during the Middle Miocene.

1. Introduction

It is generally believed that the Japanese Islands, once situated close to the Asian continent, drifted toward the Pacific Ocean when a back-arc basin, the Japan Sea, was formed [1,2]. This idea is supported by paleomagnetic direction data from Southwest Japan; Kawai et al. [3,4] reported paleomagnetic directions of Cretaceous rocks in Southwest Japan which were deflected clockwise from the north-south direction. Similar results were also observed in Paleogene rocks in the same region by Sasajima et al. [5]. This line of evidence suggests that Southwest Japan was rotated clockwise, possibly associated with opening of the Japan Sea.

Exact timing of the formation of the Japan Sea has not been established so far; magnetic anomalies on the Japan Sea floor have not been convincingly identified [6], and basal sediments were not obtained by the Deep Sea Drilling Project [7]. It was inferred, however, that the spreading of the Japan Sea basin was initiated in the late Cretaceous to Paleogene times [8–10]. Sillitoe [11] suggested, using the correlation of metallogenic events in Korea and Japan, that southward rafting of the Japanese Islands began after the end of Cretaceous, and possibly after the mid-Eocene. Recent improvement of paleomagnetic data from Southwest Japan, however, suggests that the formation of the Japan Sea was a much younger event than previously reported. Otofuji and Matsuda [12] made a paleomagnetic study on the late Cretaceous to Miocene igneous rocks from the San-in district in Southwest Japan, and observed a clockwise deflection of the paleomagnetic declination in the latest Oligocene rocks dated at 28 Ma, as well as in the older Paleogene and Cretaceous rocks. Thus, it is evident that the clockwise rotation of Southwest Japan was initiated after 28 Ma.

In order to clarify the timing of the rotation of Southwest Japan and formation of the Japan Sea, it is necessary to reassess paleomagnetic directions recorded in Miocene rocks. Although several paleomagnetic directions including anomalous ones have been reported from Miocene igneous rocks in

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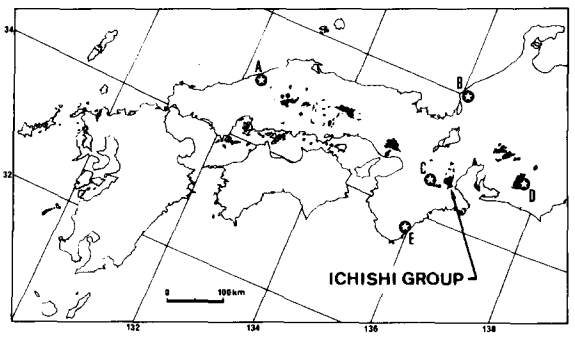


Fig. 1. Map showing location of the Ichishi Group and other discussed regions: A = Cretaceous to Middle Miocene igneous rocks along the Go River [12], B = Ito Formation in Niu Mountains [14], C = Muro Volcanics [15], D = Shidara dyke swarm [16], E = Kumano acidic rocks [17]. Base map shows distribution of the Setouchi Miocene Series.

Southwest Japan [13–17], systematic paleomagnetic studies on Miocene sedimentary sequences have not been carried out as yet. Some of the anomalous magnetic directions of the Miocene igneous rocks have tended to be interpreted as records of the geomagnetic field during a polarity transition or excursion [13,15,16]. A paleomagnetic study of a thick sedimentary sequence which covers a sufficient time interval may clarify whether the deflected magnetic directions were due to an anomalous geomagnetic field behavior of a short period, or whether they can be attributed to an averaged dipole field of which direction was deflected through a tectonic rotation. Furthermore, bedding planes of sedimentary rocks potentially guarantee appropriate tilt correction of paleomagnetic directions.

In this report, we describe results of a paleomagnetic study on a Miocene sedimentary sequence, the Ichishi Group, exposed in Mie Prefecture, Southwest Japan (Fig. 1). Using these data, we estimate a possible paleoposition of Southwest Japan at about 16 Ma.

2. Geological setting and sampling

In Southwest Japan, the distribution of Miocene sedimentary rocks sporadically extends east and west. Lithological and paleontological evidence suggests that these rocks were formed in a shallow-marine environment, similar to that of the present Setouchi Inland Sea in Southwest Japan. They are collectively called the Setouchi Miocene Series [18]. The Ichishi Group is a typical sequence of the Setouchi Miocene Series. Its thickness reaches about 1000 m, which is unusual among the Setouchi Series.

The stratigraphy of the Ichishi Group is summarized in Fig. 2. It is divided into three lithostra-

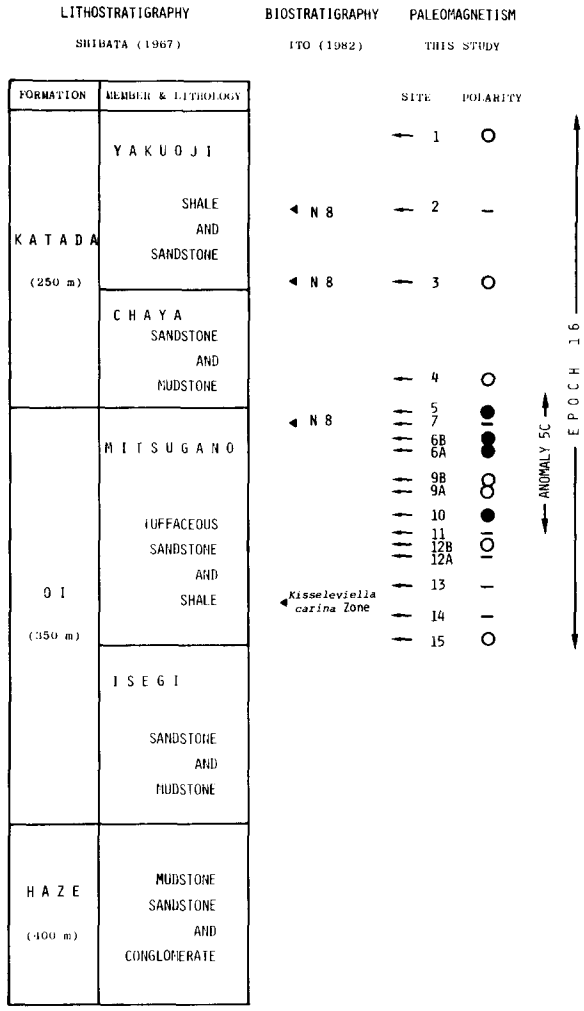


Fig. 2. Summary of lithostratigraphy [19] and biostratigraphy [20] of the Miocene Ichishi Group, and results of paleomagnetic polarity stratigraphy. Solid and open circles represent normal and reversed polarity, respectively. Dashes indicate sites that were rejected due to instability or excessive angular dispersion of the magnetic directions.

tigraphic units: Haze, Oi and Katada Formations in ascending order [19]. The Haze Formation unconformably covers the basement Ryoke Metamorphic Complex, and is mainly composed of coarse-grained sandstones and conglomerates. The Oi Formation consists of the Isegi sandstone and mudstone member in the lower part and the Mitsugano tuffaceous sandstone and mudstone member in the upper part. The Mitsugano member intercalates several layers of water-laid tuff, most of which are composed of fine-grained volcanic ashes. The Katada Formation is also subdivided into two members: the Chaya sandstone and mudstone member, and the Yakuoji alternation member.

Recently, the occurrence of foraminifera, diatoms and radiolaria was reported from 4 horizons in the upper part of the Ichishi Group [20] (Fig. 2). In a diatom assemblage from the lower part of the Mitsugano member, *Kissleviella carina* was dominant, and *Actinocyclus ingens* was absent. Therefore, this horizon is correlated to the *Kissleviella carina* zone of the diatom biostratigraphy [21]. Planktonic foraminifera assemblages including *Praeorbulina glomerosa* and *Globigerinoides sicanus* were found from the uppermost Mitsugano member and the Yakuoji member, thus correlating these horizons to Zone N8 of the planktonic foraminifera biostratigraphy [22]. These micropaleontological data indicate that the upper part of the Ichishi Group is assigned to the late Early Miocene.

Paleomagnetic sampling was carried out along the route where the biostratigraphic work [20] was previously performed. Samples were collected at a total of 17 sites. These sites were collected from mud or tuff layers of the Katada Formation and the Mitsugano member of the Oi Formation (Fig. 2). In the sampling area, structural attitude of the Ichishi Group was simple; the strata gently dip less than 10° generally to the east, and tectonic disturbance or slumping structure was not observed. In this study, a site is defined as a single bed of mudstone or tuff which has a thickness less than 1 m and is exposed in a continuous outcrop about 5 to 10 m wide. At each site, six samples were collected from a stratigraphic interval of about 30 cm thick. The samples were independently oriented by a magnetic compass. Strike and dip of the strata were also measured for tilt correction of the remanent magnetic direction. Each sample was cut into two 2-cm cubes in the laboratory for subsequent treatment.

3. Laboratory procedure and results

Remanent magnetization of specimens was measured with a cryogenic magnetometer (ScT C-112) at Kyoto University. Magnetic directions

TABLE 1

Paleomagnetic site mean directions from the Ichishi Group

Site	T	D_m	I_m	N	A_{95}	k	Polarity	PHI	LMD
1	250	−139.9	−32.7	6	41.4	3.6	R	50.5	−118.6
3	250	−126.8	−31.6	6	13.4	25.9	R	39.6	−127.2
4	150	−154.1	−37.3	5	25.5	9.9	R	63.3	−109.0
5	150	34.0	41.7	4	30.8	9.9	N	58.6	−122.7
6B	150	99.4	69.8	5	23.0	12.0	N	22.3	175.6
6A	250	44.2	37.9	5	22.0	13.0	N	49.0	−126.3
9B	250	−96.3	−63.0	5	9.2	70.4	R	27.6	−170.3
9A	250	−146.2	−56.8	3	23.1	29.7	R	62.6	−149.0
10	250	13.0	61.1	5	7.8	98.4	N	77.4	−173.5
12B	450	−153.1	−47.5	6	6.2	118.1	R	66.3	−126.0
15	250	−107.4	−34.4	4	39.9	6.3	R	24.6	−140.2

T = temperature of thermal demagnetization in °C; D_m, I_m = mean declination and inclination in degrees; N = number of specimens; A_{95} = radius of 95% confidence circle in degrees; k = precision parameter; PHI, LMD: latitude and longitude of virtual geomagnetic pole position (north-seeking pole).

were corrected for the field orientation of the samples, for the tilt of the strata, and for the local declination of the present geomagnetic field (6.3°W). Stability of remanent magnetization was investigated by alternating field (AF) and thermal demagnetization, to which two specimens from an independently oriented sample were respectively subjected. Typical examples of magnetic behavior upon stepwise AF and thermal demagnetizations are shown in Fig. 3 using orthogonal vector plots.

Most specimens showed remarkable changes in magnetic direction on beginning of the progressive AF demagnetization, such as in peak fields of 50 or 100 Oe. The soft component of remanent magnetization which was cleaned by the AF treatment can be interpreted as a secondary overprint of viscous remanent magnetization (VRM). After demagnetizations in peak AF of 100 Oe, remanent magnetization of most sites showed a general grouping in direction. Specimens from six sites, however, continued exhibiting erratic changes in remanent direction, and significant grouping within the site was not obtained through further treatments of demagnetization.

The soft VRM component was also erased by thermal demagnetization at temperatures above 150°C. Further thermal treatments revealed that blocking temperature is distributed between 150 and 450°C, where the magnetization vectors were observed to decay toward the origin. It was also confirmed that the secondary overprint of chemical remanent magnetization (CRM), such as car-

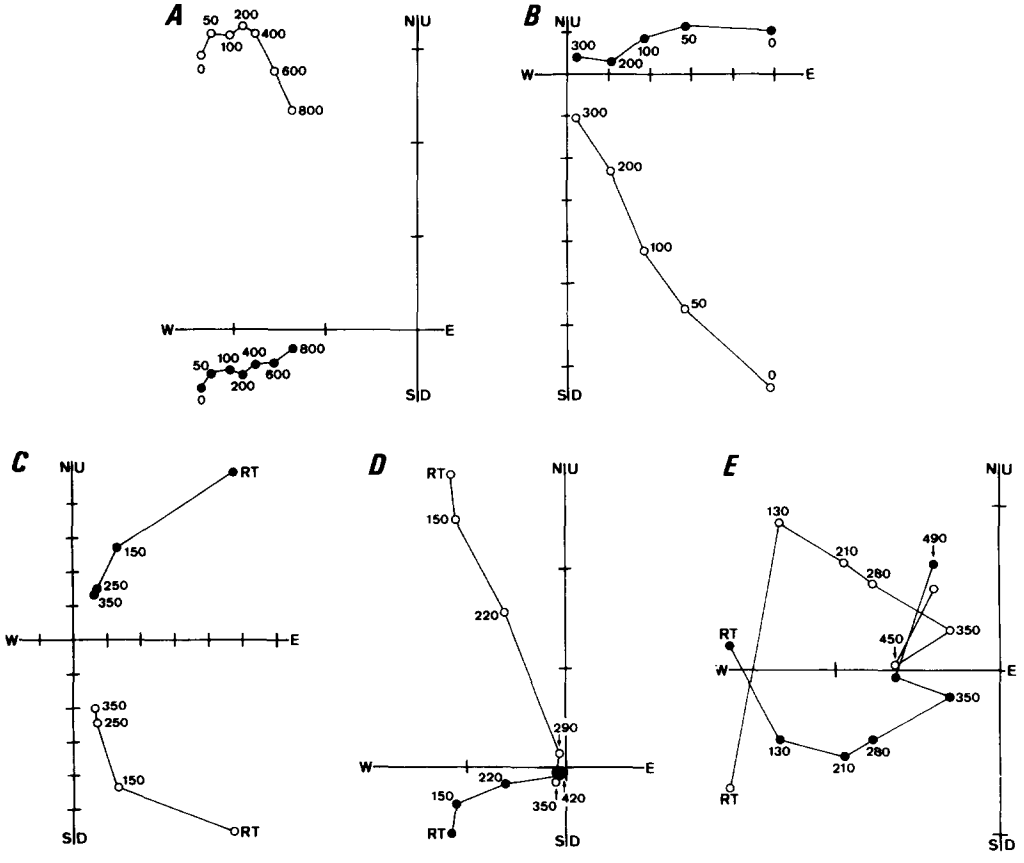


Fig. 3. Behavior of remanent magnetization in response to progressive AF (A, B) and thermal (C, D, E) demagnetization. Solid and open circles represent horizontal and vertical components of magnetization, respectively. Numerals show peak AF values in oersted (A, B) and temperatures of thermal demagnetization in °C (C, D, E). One division of each scale is 1×10^{-7} emu/cm^3.

ried by goethite, was scarce in most of the samples. Remanent magnetization of some volcanic ash samples was able to be decreased to less than 3% of the initial intensity after thermal demagnetization. Magnetization of other specimens, however, could not be removed perfectly; after thermal treatments at 350°C or higher, most specimens acquired quite unstable magnetizations with intensities exceeding their initial value. This often prevented a determination of the high-temperature component.

Site mean directions of remanent magnetization are listed in Table 1. The site mean was selected from the results of thermal demagnetizations at 150–450°C on the basis of the greatest value of the precision parameter (k; [23]). Because several specimens were broken into fragments during the thermal treatment, the number of specimens is fewer than 6 in some sites. Six sites which showed highly dispersed magnetic directions ($k < 3$) were omitted from the table.

The site mean directions are grouped into two separate clusters in antipodal directions (Fig. 4). These two directions are regarded as of normal and reversed polarity, while the declinations are deflected to the east and west, respectively. We believe that these paleomagnetic data represent the direction of the ancient geomagnetic field at the time of accumulation and subsequent compaction of the sediments. This statement is supported by the following facts: (1) magnetic stability was examined using the method of AF and thermal demagnetization; (2) magnetic directions were corrected for the tilt ($< 10°$) of the strata; (3) remanent magnetizations were classified into normal and reversed polarity.

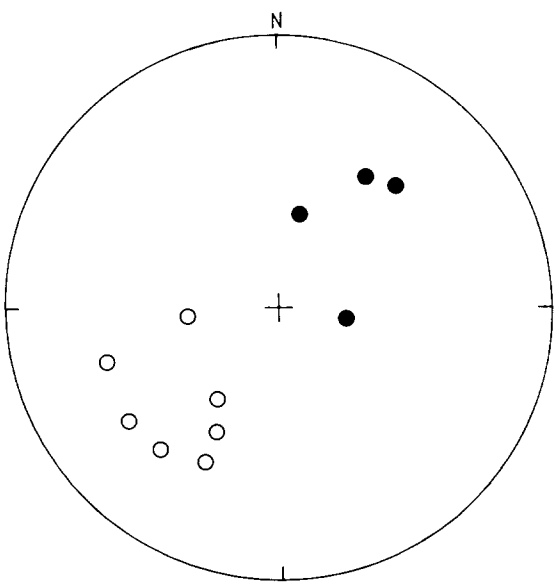


Fig. 4. Site mean directions of remanent magnetization, after tilt correction.

4. Discussion

The stratigraphic distribution of normal and reversed polarity sites is represented in Fig. 2. The upper Ichishi Group is divided into at least 5 magnetozones; the upper and lower parts of the studied section are magnetized with reversed polarity, and two normal polarity zones exist in the middle part. The biostrtatigraphic correlation of the Ichishi Group to the late Early Miocene including Blow's Zone N8 [20] allows us to assign the polarity sequence to a conventional polarity time scale. Opdyke et al. [24] obtained the result that the *Orbulina* datum, which corresponds to the N9/N8 boundary (the Early/Middle Miocene boundary [25]), occurs at the Epoch 15/16 boundary of geomagnetic polarity history. We interpret, therefore, that the Ichishi Group is correlated to the reversed polarity interval, Epoch 16, which is dated from 17.6 to 15.2 Ma [26]. The normal polarity zones found in the middle part of the section are assigned to normal polarity events, which correspond to anomaly 5C of the marine magnetic anomaly timescale, dated at 17.0–16.2 Ma [26].

When reversed polarity directions are inverted to the normal ones, site mean directions from the 11 horizons (Table 1) yield the mean declination: $D = 45.1°$ and mean inclination: $I = 48.8°$ with the radius of 95% confidence circle: $A_{95} = 11.9°$. While the mean inclination value is not significantly different from that of the present geomag-

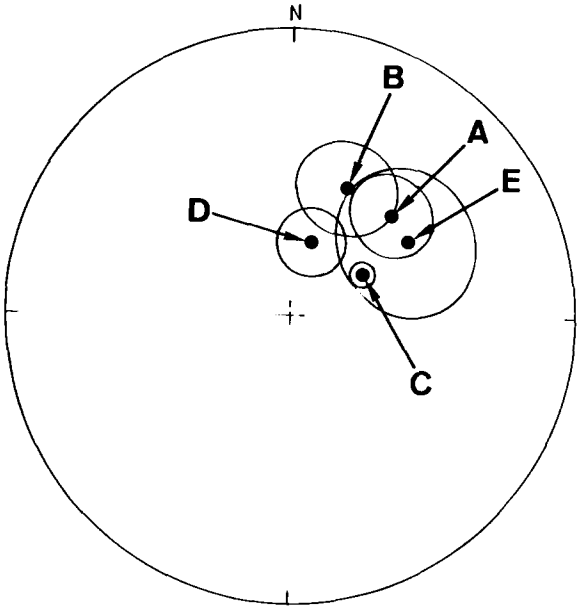


Fig. 5. Paleomagnetic directions of Miocene rocks dated at 15–17 Ma in Southwest Japan. *A* = Ichishi Group (this study), *B* = Ito Formation in Niu Mountains [14], *C* = Muro volcanics [15], *D* = Shidara dyke swarm [16], *E* = Kumano acidic rocks [17].

netic field (49°) or the geocentric axial dipole field (54°), the mean declination is easterly deflected from the present north. This deflection exceeds the uncertainty of declination: Δ_{95} ($= \sin^{-1}(\sin A_{95}/\cos I)$, [27]) = 18.2°. The more rigorous estimate for the 95% confidence limit of declination is calculated at 14.4° by using the approximation method of Demarest [28]. The magnetostratigraphic results mentioned above indicate that our paleomagnetic data set is a record of the geomagnetic field which covers a sufficiently long period, including both normal and reversed polarity intervals, so that the mean direction is regarded as representing the direction of the averaged dipole field during Epoch 16.

The deflected magnetic direction of the Ichishi Group is not attributed to polar wander on a global scale, but explained by a local tectonic rotation of the area. The late Tertiary paleomagnetic poles determined from Eurasia [29] almost coincide with the present geographic pole. This fact implies that the Neogene motion of the Asian continent relative to the pole was of a negligible amount. It may thus be concluded that the area including the Ichishi Group was rotated relative to the Asian continent after 16 Ma.

TABLE 2

Paleomagnetic data of 15–17 Ma from Southwest Japan

Locality	Material	D_m	I_m	A_{95}	Age [a] (Ma)	Reference
Ichishi 34.7°N, 136.4°E	mud tuff	45.1	48.8	11.9	15.2–17.6	this study
Niu Mts. 36.0°N, 136.1°E	welded tuff lava	23.7	49.9	14.0	16.0 (FT) 18.6 (FT)	[14]
Muro 34.5°N, 136.0°E	welded tuff	60.6	66.1	3.7	15.0 (FT) 17.5 (K-Ar)	[15,31]
Shidara 35.1°N, 137.6°E	dyke rock	15.9	68.3	9.8	15.1 (K-Ar)	[16]
Kumano 33.8°N, 136.0°E	lava welded tuff intrusion	57.5	49.7	20.6	14.4 (K-Ar) 15.0 (FT) 15.1 (FT)	[17,32]

D_m, I_m = mean declination and inclination in degrees; A_{95}: radius of 95% confidence circle in degrees.

[a] FT: Fission-track age.

Note: Each mean direction was calculated by the present authors. Numeric data of Muro and Kumano were provided by M. Torii and T. Tagami, respectively.

Fig. 5 shows paleomagnetic directions of the Miocene rocks in Southwest Japan which were dated at about 17 to 15 Ma (Table 2). As shown in the figure, the declinations, which were converted to the normal direction, are all deflected easterly from 15° to 60°. Some of these deflected magnetic directions were interpreted to represent anomalous behavior of the geomagnetic field during a polarity transition or excursion [13,15,16]. In the Ichishi Group, however, deflection of the declination was observed not on a restricted horizon but through the sequence which is estimated to cover almost 2.4 Ma (Epoch 16). In other words, the deflected magnetic direction of the Ichishi Group can be regarded as a record of the averaged geomagnetic field direction, which is acceptable as a tectonic indicator. It seems possible, therefore, that the clockwise deflection of magnetic directions observed in Miocene rocks other than the Ichishi Group were also caused by tectonic rotation.

Otofuji and Matsuda [12] showed that Southwest Japan was rotated clockwise through 58° ± 14° (Δ_{95}) between 28 Ma and 12 Ma. The easterly deflected declinations observed in the Ichishi Group and other Miocene rocks in Southwest Japan suggest that the clockwise rotation occurred later than about 15 Ma. According to the mean declination value of the Ichishi Group, the rotation angle after 16 Ma is estimated 45° ± 18° (Δ_{95}), which is not significantly different from the rotation angle after 28 Ma. This fact suggests that Southwest Japan was subjected to no detectable rotation between 28 Ma and 16 Ma. It is concluded, therefore, that the majority of the clockwise rotation occurred between 16 Ma and 12 Ma.

The clockwise rotation of Southwest Japan can be attributed to its drift from the Asian continent associated with the formation of the Japan Sea. The rotation might also be explained by other tectonic models than the opening of the Japan Sea; the easterly deflected declinations could be caused by rotations of small land blocks in a right-lateral shear zone, or by the convergence of the western part of Southwest Japan toward the Korean Peninsula. These models are, however, hardly acceptable, because geologic evidence proposing the occurrence of such large scale shear motions or convergence to the Korean Peninsula

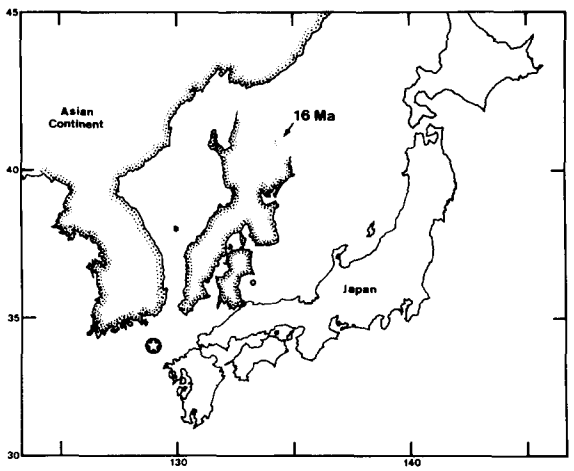


Fig. 6. A possible paleoposition of Southwest Japan at 16 Ma, depicted in Mercator projection. Southwest Japan was situated near the Asian continent, before the clockwise rotation through 45°. Pivot pole (star) is located at 34°N, 129°E [12]. Gap between the Asian continent and Southwest Japan might be filled with the Yamato Ridge and the North Korean Rise.

in the Neogene does not exist in and around Southwest Japan. Instead, the presence of a back-arc basin, the Japan Sea, implies that divergence has occurred between the Japanese Islands and the Asian continent. Thus, the hypothesis that the rotation of Southwest Japan was a consequence of the opening of the Japan Sea is most favored.

The timing of the clockwise rotation of Southwest Japan suggests that the Japan Sea opened during the Middle Miocene. Fig. 6 shows a possible paleoreconstruction at 16 Ma, where Southwest Japan was "unrotated" through 45° around the pivot at 34°N, 129°E determined by Otofuji and Matsuda [12]. It is supposed that Southwest Japan had been situated adjacent to the Korean Peninsula until 16 Ma, and then migrated to the present position before 12 Ma accompanying back-arc spreading of the Japan Sea. Based on lithostratigraphic and biostratigraphic data from several sections exposed on the Japan Sea coast, Ingle [30] summarized the Neogene history of the area as follows: (1) deposition of andesitic volcanic materials (so-called "green tuffs") and non-marine sediments in the late Oligocene and Early Miocene, (2) marine transgression in the latest Early Miocene (Nishikurosawa Stage), and (3) subsidence in the Middle and Late Miocene. Thus, the stratigraphic

evidence around the Japan Sea also suggests its opening in the Middle Miocene.

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